

06

CRYPTOGRAPHY

5 LESSONS

PHASE 2 · CORE SECURITY

Cryptography & Public Key Infrastructure

Symmetric and asymmetric encryption, hashing, digital signatures, TLS, and PKI – the math that protects everything.

01

Cryptography Foundations

BEGINNER

CONCEPT

30 min →

Encryption vs encoding, symmetric vs asymmetric, confidentiality vs i...

02

Hashing, MACs & Data Integrity

INTERMEDIATE

LAB

30 min →

Hash functions, password storage, HMAC, and how integrity checks catc...

03

Symmetric & Asymmetric Algorithms

INTERMEDIATE

CONCEPT

40 min →

AES, ChaCha20, RSA, ECC – understand the algorithms, their key sizes,...

04

Digital Certificates & PKI

INTERMEDIATE

CONCEPT

40 min →

Certificate authorities, chains of trust, X.509, and how browsers and...

05

TLS in Practice: Handshakes & Attacks

INTERMEDIATE

LAB

40 min →

The full TLS 1.3 handshake, cipher negotiation, forward secrecy, and ...

Identity & Access Management →

